

FIRST FOLD SHOULD BE HORIZONTAL REMAINING VERTICAL

MYCOKEM CAPSULES 250 mg
Mycophenolate Mofetil Capsules USP

NAME OF PRODUCT
MYCOKEM CAPSULES 250 MG

DESCRIPTION AND COMPOSITION:

MYCOKEM CAPSULES 250 MG:

White to off white granular powder filled in size "11" blue coloured cap, brown coloured body hard gelatin capsules imprinted "266" on body with black ink.

Each hard gelatin capsule contains:

Mycophenolate Mofetil Ph. Eur. 250 mg

PHARMACODYNAMICS

Pharmacotherapeutic group: immunosuppressive agents ATC code L04AA06

Mycophenolate mofetil is the 2-morpholinoethyl ester of MPA. MPA is a potent, selective, uncompetitive and reversible inhibitor of inosine monophosphate dehydrogenase, and therefore inhibits the de novo pathway of guanosine nucleotide synthesis without incorporation into DNA. Because T- and B-lymphocytes are critically dependent for their proliferation on de novo synthesis of purines whereas other cell types can utilise salvage pathways, MPA has more potent cytostatic effects on lymphocytes than on other cells.

PHARMACOKINETICS

Absorption

Following oral administration, mycophenolate mofetil undergoes rapid and extensive absorption and complete presystemic metabolism to the active metabolite, MPA. As evidenced by suppression of acute rejection following renal transplantation, the immunosuppressant activity of Mycokem is correlated with MPA concentration. The mean bioavailability of oral mycophenolate mofetil, based on MPAAUC, is 94% relative to IV mycophenolate mofetil. Mycophenolate mofetil is not measurable systemically in plasma following oral administration.

Distribution

As a result of enterohepatic recirculation, secondary increases in plasma MPA concentration are usually observed at approximately 6 – 12 hours post-dose. A reduction in the AUC of MPA of approximately 40% is associated with the co-administration of cholestyramine (4 g TID), indicating that there is a significant amount of enterohepatic recirculation. MPA at clinically relevant concentrations is 97% bound to plasma albumin.

Biotransformation

MPA is metabolised principally by glucuronyl transferase (isoform UGT1A9) to form the inactive phenolic glucuronide of MPA (MPAG). In vivo, MPAG is converted back to free MPA via enterohepatic recirculation. A minor acylglucuronide (AcMPAG) is also formed. AcMPAG is pharmacologically active and is suspected to be responsible for some of MMF's side effects (diarrhoea, leucopenia).

Elimination

A negligible amount of substance is excreted as MPA (< 1 % of dose) in the urine. Oral administration of radiolabelled mycophenolate mofetil results in complete recovery of the administered dose with 93% of the administered dose recovered in the urine and 6% recovered in the faeces. Most (about 87%) of the administered dose is excreted in the urine as MPAG.

At clinically encountered concentrations, MPA and MPAG are not removed by haemodialysis. However, at high MPAG plasma concentrations (> 100µg/mL), small amounts of MPAG are removed. By interfering with enterohepatic circulation of the drug, bile acid sequestrants such as cholestyramine, reduce MPAAUC.

MPA's disposition depends on several transporters. Organic anion-transporting polypeptides (OATPs) and multidrug resistance-associated protein 2 (MRP2) are involved in MPA's disposition; OATP isoforms, MRP2 and breast cancer resistance protein (BCRP) are transporters associated with the glucuronides' biliary excretion. Multidrug resistance protein 1 (MDR1) is also able to transport MPA, but its contribution seems to be confined to the absorption process. In the kidney MPA and its metabolites potentially interact with renal organic anion transporters.

Special populations

Renal impairment

Dose adjustment may be required in patients of renal impairment

Delayed renal graft function

There may be a transient increase in the free fraction and concentration of plasma MPA in patients with delayed renal graft function. Dose adjustment of Mycokem does not appear to be necessary.

Hepatic impairment

Effects of hepatic disease on glucuronidation of MPA probably depend on the particular disease. However, hepatic disease with predominantly biliary damage, such as primary biliary cirrhosis, may show a different effect.

Elderly

Pharmacokinetic behaviour of Mycokem in the elderly (≥ 65 years) has not been formally evaluated.

INDICATIONS

- Mycokem capsules and tablets are indicated for the prophylaxis of acute organ rejection in patients receiving allogeneic renal transplants.
- Mycokem capsules and tablets are indicated for the prophylaxis of acute organ rejection in patients receiving allogeneic cardiac transplants.
- Mycokem capsules and tablets are indicated for the prophylaxis of acute organ rejection in patients receiving allogeneic hepatic transplants.
- Mycokem capsules and tablets used concomitantly with ciclosporin and corticosteroids.
- Mycokem capsules and tablets indicated for induction and maintenance treatment of lupus nephritis. Mycokem capsules and tablets should be used concomitantly with corticosteroids.

RECOMMENDED DOSE

Standard dosage for prophylaxis of renal rejection:

A dose of 1 g administered twice a day (daily dose of 2g) is recommended for use in renal transplant patients. Although a dose of 1.5g administered twice daily (daily dose of 3g) was used in clinical trials and was shown to be safe and effective, no efficacy advantage could be established for renal transplant patients. Patients receiving 2 g per day of Mycokem capsules demonstrated an overall better safety profile compared to patients receiving 3g per day of Mycokem capsules.

Standard dosage for prophylaxis of cardiac rejection:

A dose of 1.5g administered twice a day (daily dose of 3g) is recommended for use in cardiac transplant patients.

Standard dosage for prophylaxis of hepatic rejection:

The initial dose of Mycokem capsules should be given as soon as possible following transplantation. The recommended dose in hepatic transplant patients is 1.5g administered twice daily (daily dose of 3 g).

Oral administration

The initial dose of Mycokem capsules should be given as soon as possible following renal, cardiac and hepatic transplantation.

Induction and maintenance treatment of lupust nephritis

The recommended dose is 1g administered twice a day (daily dose of 2g).

Special dosage instructions:

Patients with neutropenia:

If neutropenia develops (absolute neutrophil count < 1.3 x 10⁹/µl), dosing with Mycokem capsules should be interrupted or the dose reduced.

Geriatric Use:

The recommended dose of 1g b.i.d. for renal transplant patients and 1.5g b.i.d. for cardiac or hepatic transplant patients are appropriate for elderly patients.

Renal Impairment:

Patients with severe renal impairment

In renal transplant patients with severe chronic renal impairment (glomerular filtration rate <25ml/min/1.73m²), outside of the immediate post-transplant period or after treatment of acute or refractory rejection, doses greater than 1g administered twice a day should be avoided. No data are available for cardiac or hepatic transplant patients with severe chronic renal impairment.

Patients with delayed renal graft function post-transplant

No dose adjustments are needed in patients experiencing delayed renal graft function post-operatively.

Hepatic Impairment:

No dose adjustments are needed for renal transplant patients with severe hepatic parenchymal disease.

MODE OF ADMINISTRATION

Oral route of administration

CONTRAINDICATIONS

- Mycokem is contraindicated during pregnancy due to its mutagenic and teratogenic potential (see Use in Special Populations: Pregnancy).
- Mycokem is contraindicated in women of childbearing potential not using highly effective contraceptive methods (see Use in Special Populations: Pregnancy).
- Mycokem is contraindicated in women who are breastfeeding (see Use in Special Populations: Breastfeeding).
- Mycokem should not be given to patients with hypersensitivity to mycophenolate mofetil, mycophenolic acid or to any of the excipients. Hypersensitivity reactions to Mycokem have been observed

WARNINGS AND PRECAUTIONS

Neoplasms

Patients receiving immunosuppressive regimens including combinations of medicinal products, including Mycokem, are at increased risk of developing lymphomas and other malignancies, particularly of the skin. The risk appears to be related to the intensity and duration of immunosuppression rather than to the use of any specific agent. As general advice to minimise the risk for skin cancer, exposure to sunlight and UV light should be limited by wearing protective clothing and using a sunscreen with a high protection factor. Patients receiving Mycokem should be instructed to report immediately any evidence of infection, unexpected bruising, bleeding or any other manifestation of bone marrow depression.

Patients treated with immunosuppressants, including Mycokem, are at increased risk for opportunistic infections (bacterial, fungal, viral and protozoal), fatal infections and sepsis. Such infections include latent viral reactivation, such as hepatitis B or hepatitis C reactivation and infections caused by polyomaviruses (BK virus associated nephropathy, JC virus associated progressive multifocal leukoencephalopathy PML). Cases of hepatitis due to reactivation of hepatitis B or hepatitis C have been reported in carrier patients treated with immunosuppressants. These infections are often related to a high total immunosuppressive burden and may lead to serious or fatal conditions that physicians should consider in the differential diagnosis in immunosuppressed patients with deteriorating renal function or neurological symptoms. Patients on Mycokem who develop recurrent infections should have their serum immunoglobulins measured. In cases of sustained, clinically relevant hypogammaglobulinaemia, appropriate clinical action should be considered taking into account the potent cytostatic effects that mycophenolic acid has on T- and B-lymphocytes.

Blood and immune system

Patients receiving Mycokem should be monitored for neutropenia, which may be related to Mycokem itself, concomitant medications, viral infections, or some combination of these causes. Patients taking Mycokem should have complete blood counts weekly during the first month, twice monthly for the second and third months of treatment, then monthly through the first year. If neutropenia develops (absolute neutrophil count < 1.3 x 10⁹/µl), it may be appropriate to interrupt or discontinue Mycokem.

Cases of pure red cell aplasia (PRCA) have been reported in patients treated with Mycokem in combination with other immunosuppressants. The mechanism for mycophenolate mofetil induced PRCA is unknown. PRCA may resolve with dose reduction or cessation of Mycokem therapy.

Changes to Mycokem therapy should only be undertaken under appropriate supervision in transplant recipients in order to minimise the risk of graft rejection. Patients receiving Mycokem should be instructed to report immediately any evidence of infection, unexpected bruising, bleeding or any other manifestation of bone marrow depression.

Patients should be advised that during treatment with Mycokem, vaccinations may be less effective and the use of live attenuated vaccines should be avoided. Influenza vaccination may be of value. Prescribers should refer to national guidelines for influenza vaccination.

Gastro-intestinal

Mycokem has been associated with an increased incidence of digestive system adverse events, including infrequent cases of gastrointestinal tract ulceration, haemorrhage and perforation. Mycokem should be administered with caution in patients with active serious digestive system disease

Mycokem is an IMPDH (inosine monophosphate dehydrogenase) inhibitor. On theoretical grounds, therefore, it should be avoided in patients with rare hereditary deficiency of hypoxanthine-guanine phosphoribosyl-transferase (HGPRT) such as Lesch-Nyhan and Kelley-Seegmiller syndrome.

Interactions

Caution should be exercised when switching combination therapy from regimens containing immunosuppressants, which interfere with MPA enterohepatic recirculation e.g. ciclosporin to others devoid of this effect e.g. sirolimus, belatacept, or vice versa, as this might result in changes of MPA exposure. Drugs of other classes which interfere with MPA's enterohepatic cycle e.g. cholestyramine, should be used with caution due to their potential to reduce the plasma level and efficacy of Mycokem.

It is recommended that Mycokem should not be administered concomitantly with azathioprine because such concomitant administration has not been studied. In view of the significant reduction in the AUC of MPA by cholestyramine, caution should be used in the concomitant administration of Mycokem with medicinal products that interfere with enterohepatic recirculation because of the potential to reduce the efficacy of Mycokem. The risk: benefit of mycophenolate mofetil in combination with tacrolimus or sirolimus has not been established

Special populations

Elderly patients may be at an increased risk of adverse events such as certain infections (including cytomegalovirus tissue invasive disease) and possibly gastrointestinal haemorrhage and pulmonary oedema, compared with younger individuals. Immunosuppressed patients are at increased risk for opportunistic infections, including activation of latent viral infections. These include BK virus associated nephropathy which has been observed in patients receiving immunosuppressants. These infections may lead to serious, including fatal outcomes.

Teratogenic effects

Mycophenolate is a powerful human teratogen. Therefore Mycokem is contraindicated in pregnancy unless there are no suitable alternative treatments to prevent transplant rejection. Female and male patients of reproductive potential should be made aware of the risks and follow the recommendations (e.g. contraceptive methods, pregnancy testing) prior to, during, and after therapy with Mycokem. Physicians should ensure that women and men taking mycophenolate understand the risk of harm to the baby, the need for effective contraception, and the need to immediately consult their physician if there is a possibility of pregnancy.

Contraception

Because of the genotoxic and teratogenic potential of Mycokem, women with childbearing potential should use two reliable forms of contraception simultaneously before starting Mycokem therapy, during therapy, and for six weeks after stopping the therapy; unless abstinence is the chosen method of contraception. Sexually active men are recommended to use condoms during treatment and for at least 90 days after cessation of treatment. Condom use applies for both reproductively competent and vasectomised men, because the risks associated with the transfer of seminal fluid also apply to men who have had a vasectomy. In addition, female partners of male patients treated with Mycokem are recommended to use highly effective contraception during treatment and for a total of 90 days after the last dose of Mycokem.

INTERACTIONS WITH OTHER MEDICATIONS

Interaction studies have only been performed in adults.

Aciclovir: Concurrent use of aciclovir and mycophenolate mofetil will result in higher aciclovir plasma concentrations. The changes in MPAG (the phenolic glucuronide of MPA) were minimal and are not considered clinically significant. Because MPAG plasma concentrations are increased in the presence of renal impairment, as are aciclovir concentrations, the potential exists for mycophenolate mofetil and aciclovir, or its prodrugs, e.g. valaciclovir, to compete for tubular secretion and further increases in concentrations of both substances may occur.

Antacids and proton pump inhibitors (PPIs): Concurrent use of antacids, such as magnesium and aluminium hydroxides and PPIs, including lansoprazole and pantoprazole with Mycokem may decrease the MPA (Mycophenolic acid) exposure.

Cholestyramine: Caution should be used during concomitant administration because of the potential to reduce efficacy of Mycokem.

Medicinal products that interfere with enterohepatic circulation: caution should be used with medicinal products that interfere with enterohepatic circulation because of their potential to reduce the efficacy of Mycokem.

Ciclosporin A: Ciclosporin A (CsA) pharmacokinetics are unaffected by mycophenolate mofetil. CsA interferes with MPA enterohepatic recycling, resulting in reduced MPA exposures in renal transplant patients treated with Mycokem and CsA compared with patients receiving sirolimus or belatacept and similar doses of Mycokem. Conversely, changes of MPA exposure should be expected when switching patients from CsA to one of the immunosuppressants which does not interfere with MPA's enterohepatic cycle.

Ganciclovir: It is anticipated that co-administration of oral mycophenolate and IV ganciclovir (which compete for mechanisms of renal tubular secretion) will result in increases in MPAG and ganciclovir concentration. No substantial alteration of MPA pharmacokinetics is anticipated and Mycokem dose adjustment is not required. In patients with renal impairment in which Mycokem and ganciclovir or its prodrugs, e.g. valganciclovir, are co-administered, the dose recommendations for ganciclovir should be observed and patients should be monitored carefully.

Oral contraceptives: the pharmacokinetics and pharmacodynamics of oral contraceptives were unaffected by coadministration of Mycokem. Rifampicin: in patients not also taking Ciclosporin, concomitant administration of Mycokem and rifampicin resulted in a decrease in MPA exposure. It is recommended to monitor MPA exposure levels and to adjust Mycokem doses accordingly to maintain efficacy when rifampicin is administered concomitantly.

Sevelamer: It is recommended to administer Mycokem at least one hour before or three hours after sevelamer intake to minimise the impact on the absorption of MPA.

Norflloxacin and metronidazole: The combination of norflloxacin and metronidazole together may reduce the MPA exposure when used after a single dose of Mycokem.

Ciprofloxacin and amoxicillin plus clavulanic acid: Immediately following commencement of oral ciprofloxacin or amoxicillin plus clavulanic acid in renal transplant patients, there may be reduction in pre-dose (trough) MPA concentrations. This effect tended to diminish with continued antibiotic use and to cease within a few days of their discontinuation. The change in predose level may not accurately represent changes in overall MPA exposure. Therefore, a change in the dose of Mycokem should not normally be necessary in the absence of clinical evidence of graft dysfunction. However, close clinical monitoring should be performed during the combination and shortly after antibiotic

treatment.
Tacrolimus: Concurrent use of multiple doses of Mycokem and tacrolimus in hepatic transplant patients may result in increase in concentration of tacrolimus.
Other interactions: Co-administration of probenecid with mycophenolate mofetil may raise the concentration of MPAG. Thus, other substances known to undergo renal tubular secretion may compete with MPAG, and thereby raising the plasma concentrations of MPAG or the other substance undergoing tubular secretion.
Live vaccines: Live vaccines should not be given to patients with an impaired immune response. The antibody response to other vaccines may be diminished.

PREGNANCY AND LACTATION

Pregnancy
Mycokem is contraindicated during pregnancy and in women of childbearing potential not using highly effective contraceptive methods. (see Contraindications)

Before the start of treatment, female and male patients of reproductive potential must be made aware of the increased risk of pregnancy loss and congenital malformations and must be counseled regarding pregnancy prevention, and planning. Prior to starting therapy with Mycokem, female patients of childbearing potential must have two negative serum or urine pregnancy tests with a sensitivity of at least 25 mIU/mL; The second test should be performed 8-10 days after the first one and immediately before starting [product name]. Repeat pregnancy tests should be performed during routine follow-up visits. Results of all pregnancy tests should be discussed with the patient. Patients should be instructed to consult their physician immediately should they become pregnant.

Due to the mutagenic and teratogenic potential of mycophenolate, women of child bearing potential should use two reliable forms of contraception simultaneously, including at least one highly effective method, before beginning mycophenolate therapy, during therapy, and for six weeks following discontinuation of therapy, unless abstinence is the chosen method of contraception.

Sexually active men are recommended to use condoms during treatment and for at least 90 days after cessation of treatment. Condom use applies for both reproductively competent and asectomised men, because the risks associated with the transfer of seminal fluid also apply to men who have had a vasectomy. In addition, female partners of male patients are recommended to use highly effective during treatment and for total of 90 days after the last dose of Mycokem.

Congenital malformations, including multiple malformations have been reported post-marketing in children of patients exposed to mycophenolate in combination with other immunosuppressants during pregnancy. The following malformations were most frequently reported:

- Facial malformations such as cleft lip, cleft palate, micrognathia and hypertelorism of the orbits;
 - Abnormalities of the ear (e.g. abnormally formed or absent external/middle ear) and eye (e.g. coloboma, microphthalmos);
 - Malformations of the fingers (e.g. polydactyly, syndactyly, brachydactyly);
 - Cardiac abnormalities such as atrial and ventricular septal defects;
 - Oesophageal malformations (e.g. oesophageal atresia);
 - Nervous system malformations (such as spina bifida).
- In the medical literature, malformations in children from mycophenolate exposed pregnancies have been reported in 23% to 27% of live births. For comparison, the risk of malformations is estimated at approximately 2% of live births in the overall population and at approximately 4% to 5 % in solid organ transplant patients treated with immunosuppressants other than mycophenolate.
- Cases of spontaneous abortions have also been reported in patients exposed to mycophenolate, mainly in the first trimester. In the medical literature, the risk has been reported at 45% to 49% following mycophenolate exposure, compared to a reported rate between 12 and 33% in solid organ transplant patients treated with other immunosuppressants.
- Studies in animals have shown reproductive toxicity.

Breastfeeding

Mycokem is contraindicated during breastfeeding due to the potential for serious adverse reactions in nursing infants (see Contraindications). Studies in rats have shown mycophenolate to be excreted in milk. It is not known whether this medicine is excreted in human milk.

SIDE EFFECTS

The principal adverse reactions associated with the administration of Mycokemin combination with ciclosporin and corticosteroids include diarrhoea, leucopenia, sepsis and vomiting, and there is evidence of a higher frequency of certain types of infections.

Malignancies:

Patients receiving immunosuppressive regimens involving combinations of medicinal products, including Mycokem, are at increased risk of developing lymphomas and other malignancies, particularly of the skin. Lymphoproliferative disease or lymphoma developed in patients receiving Mycokem (2 g or 3 g daily) in combination with other immunosuppressants in renal (2 g data), cardiac and hepatic transplant patients followed for at least 1 year. Non-melanoma skin carcinomas occurred in patients; other types of malignancy occurred in patients. Three-year safety data in renal and cardiac transplant patients did not reveal any unexpected changes in incidence of malignancy compared to the 1-year data. Hepatic transplant patients were followed for at least 1 year, but less than 3 years.

Opportunistic infections:

All transplant patients are at increased risk of opportunistic infections; the risk increased with total immunosuppressive load. The most common opportunistic infections in patients receiving Mycokem (2 g or 3 g daily) with other immunosuppressants in renal (2 g data), cardiac and hepatic transplant patients followed for at least 1 year were candida mucocutaneous, CMV viraemia/syndrome and Herpes simplex.

Children and adolescents (aged 2 to 18 years):

The treatment-related adverse events were more frequent in the paediatric population, particularly in children under 6 years of age, when compared to adults: diarrhoea, sepsis, leucopenia, anaemia and infection.

Elderly patients (65 years):

Elderly patients (65 years) may generally be at increased risk of adverse reactions due to immunosuppression. Elderly patients receiving Mycokem as part of a combination immunosuppressive regimen, may be at increased risk of certain infections (including cytomegalovirus tissue invasive disease) and possibly gastrointestinal haemorrhage and pulmonary oedema, compared to younger individuals.

Other adverse reactions:

Adverse Reactions, Probably or Possibly Related to Mycokem, Reported in Patients Treated with Mycokem in Renal, Cardiac and Hepatic when Used in Combination with Ciclosporin and Corticosteroids
Within the system organ classes, undesirable effects are listed under headings of frequency, using the following categories: very common; common; uncommon; rare; very rare, not known. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

System organ class		Adverse drug reactions
Infections and infestations	Very common	Sepsis, gastrointestinal candidiasis, urinary tract infection, herpes simplex, herpes zoster
	Common	Pneumonia, influenza, respiratory tract infection, respiratory moniliasis, gastrointestinal infection, candidiasis, gastroenteritis, infection, bronchitis, pharyngitis, sinusitis, fungal skin infection, skin candida, vaginal candidiasis, rhinitis
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	Very common	-
	Common	Skin cancer, benign neoplasm of skin
Blood and lymphatic system disorders	Very common	Leucopenia, thrombocytopenia, anaemia
	Common	Pancytopenia, leukocytosis
Metabolism and nutrition disorders	Very common	-
	Common	Acidosis, hyperkalaemia, hypokalaemia, hyperglycaemia, hypomagnesaemia, hypocalcaemia, hypercholesterolaemia, hyperlipidaemia, hypophosphataemia, hyperuricaemia, gout, anorexia
Psychiatric disorders	Very common	-
	Common	Agitation, confusional state, depression, anxiety, thinking abnormal, insomnia
Nervous system disorders	Very common	-
	Common	Convulsion, hypertonia, tremor, somnolence, myasthenic syndrome, dizziness, headache, paraesthesia, dysgeusia
Cardiac disorders	Very common	-
	Common	Tachycardia

Vascular disorders	Very common	-
	Common	Hypotension, hypertension, vasodilatation
Respiratory, thoracic and mediastinal disorders	Very common	-
	Common	Pleural effusion, dyspnoea, cough
Gastrointestinal disorders	Very common	Vomiting, abdominal pain, diarrhoea, nausea
	Common	Gastrointestinal haemorrhage, peritonitis, ileus, colitis, gastric ulcer, duodenal ulcer, gastritis, oesophagitis, stomatitis, constipation, dyspepsia, flatulence, eructation
Hepatobiliary disorders	Very common	-
	Common	Hepatitis, jaundice, hyperbilirubinaemia
Skin and subcutaneous tissue disorders	Very common	-
	Common	Skin hypertrophy, rash, acne, alopecia,
Musculoskeletal and connective Tissue disorders	Very common	-
	Common	Arthralgia
Renal and urinary disorders	Very common	-
	Common	Renal impairment
General disorders and administration site conditions	Very common	-
	Common	Oedema, pyrexia, chills, pain, malaise, asthenia,
	Uncommon	de novo purine synthesis inhibitors-associated acute inflammatory syndrome
Investigations	Very common	-
	Common	Hepatic enzyme increased, blood creatinine increased, blood lactate dehydrogenase increased, blood urea increased, blood alkaline phosphatase increased, weight decreased

The following undesirable effects cover adverse reactions from post-marketing experience:

Additional adverse reactions reported during post-marketing are described below.

Gastrointestinal: gingival hyperplasia, colitis including cytomegalovirus colitis, pancreatitis, and intestinal villous atrophy.

Disorders related to immunosuppression: serious life-threatening infections including meningitis, endocarditis, tuberculosis and atypical mycobacterial infection. Cases of BK virus associated nephropathy, as well as cases of JC virus associated progressive multifocal leucoencephalopathy (PML), have been reported in patients treated with immunosuppressants, including Mycokem capsules. Agranulocytosis and neutropenia have been reported; therefore, regular monitoring of patients taking Mycokemis advised. There have been reports of aplastic anaemia and bone marrow depression in patients treated with Mycokem, some of which have been fatal.

Blood and lymphatic system disorder: Cases of pure red cell aplasia (PRCA) have been reported in patients treated with Mycokem Isolated cases of abnormal neutrophil morphology, including the acquired Pelger-Huet anomaly, have been observed in patients treated with Mycokem. These changes are not associated with impaired neutrophil function. These changes may suggest a 'left shift' in the maturity of neutrophils in haematological investigations, which may be mistakenly interpreted as a sign of infection in immunosuppressed patients such as those that receive Mycokem.

Hypersensitivity: Hypersensitivity reactions, including angioneurotic oedema and anaphylactic reaction, have been reported.

Respiratory, thoracic and mediastinal disorders: There have been isolated reports of interstitial lung disease and pulmonary fibrosis in patients treated with Mycokem in combination with other immunosuppressants, some of which have been fatal.

Congenital Disorders: Congenital malformations have been reported post-marketing in children of patients exposed to mycophenolate in combination with other immunosuppressants during pregnancy (see Use in Pregnancy).

Pregnancy, Puerperium and Perinatal Conditions: Cases of spontaneous abortions mainly in the first trimester in patients exposed to mycophenolate have been reported (see Use in Pregnancy).

General disorders and administration site conditions

De novo purine synthesis inhibitors-associated acute inflammatory syndrome has been described from post-marketing experience as a paradoxical proinflammatory reaction associated with mycophenolate mofetil and mycophenolic acid, characterised by fever, arthralgia, arthritis, muscle pain and elevated inflammatory markers. Literature case reports showed rapid improvement following discontinuation of the medicinal product.

SYMPTOMS AND TREATMENT OF OVERDOSE

In many of these cases, no adverse events were reported. In those overdose cases in which adverse events were reported, the events fall within the known safety profile of the medicinal product. It is expected that an overdose of mycophenolate mofetil could possibly result in oversuppression of the immune system and increase susceptibility to infections and bone marrow suppression. If neutropenia develops, dosing with Mycokem should be interrupted or the dose reduced. Haemodialysis would not be expected to remove clinically significant amounts of MPA or MPAG. Bile acid sequestrants, such as cholestyramine, can remove MPA by decreasing the enterohepatic re-circulation of the drug.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Mycophenolate has moderate influence on the ability to drive and use machines. It may cause somnolence, confusion, dizziness, tremor or hypotension, and therefore patients are advised to use caution when driving or using machines.

STORAGE CONDITION

Store at temperatures not exceeding 30°C in a dry place and protect from light.

SHELF LIFE

36 months from date of manufacture.

PRESENTATION

Mycokem Capsules 250 mg are packed in blister pack of 10 capsules.
Available pack sizes: Blister pack of 1 x 10 & 10 x 10 Capsules.

 Manufactured by:
ALKEM LABORATORIES LIMITED
Village Thana, Baddi, Dist. Solan,
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Product Registration Holder:

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DATE OF REVISION OF THIS LEAFLET

November 2022.